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Disrupted Physiological Coregulation in Youth at Clinical High Risk for Psychosis: Insights From a Dyadic Interaction Study

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Interpersonal difficulties have long been implicated in psychopathology. However, we know quite little about how social (dis-)connection unfolds at the physiological level in real time in clinical populations, including among youth at clinical high risk for psychosis (CHR). The present laboratory-based dyadic interaction study examined physiological coregulation in 70 dyads across 137 participants (32 CHR youth–caregiver dyads; 38 healthy youth–caregiver dyads) and linked coregulation with clinical symptoms—concurrently and prospectively. This study was based in a midsized Midwestern city and recruited from the broader community and from mental health clinics. Data were collected from 2018 to 2022. Dyads engaged in 10-min neutral, conflict, and pleasant conversations while their autonomic physiology was continuously monitored. In conflict and neutral conversations, CHR youth exhibited contrarian physiological coregulation (i.e., slowing heart rate in response to caregiver's escalating heart rate and vice versa). Contrarian coregulation was associated with elevated risk for psychosis, was linked to greater baseline psychosis symptomatology, and prospectively predicted increases in symptoms 1 year later. These findings document alterations in physiological coregulation between CHR individuals and their caregivers, highlight their relevance for clinical symptomatology, suggest novel avenues for relationship-focused treatments, and contribute to a biologically grounded science of social connection.

Keywords: social connection, physiological coregulation, dyadic interactions, clinical high risk for psychosis

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Social connection plays a critical role in well-being, health, and even longevity (e.g., Haase, 2023; Holt-Lunstad et al., 2010; Levenson, 2024; Wells et al., 2022). Interpersonal dysfunction, in turn, has long been implicated in psychopathology (e.g., Sullivan, 1953; Wright et al., 2023), and interest in the biological underpinnings of social (dis-)connection has been on the rise (e.g., Dimascio et al., 1957; Eisenberger, 2012; Levenson & Gottman, 1983; E. S. Lunkenheimer et al., 2011; Timmons et al., 2015; West & Mendes, 2023). However, we know quite little about how social disconnection unfolds *in real time at a physiological level* in clinical at-risk populations. The present dyadic interaction study explored whether alterations in physiological coregulation bear clinical relevance for youth at clinical high risk (CHR) for psychosis—a population with marked social difficulties (Cornblatt et al., 2012; Kimhy et al., 2016; Lincoln et al., 2017; Yee et al., 2021). To our knowledge, the present study was the first to examine physiological coregulation in CHR youth–caregiver dyads and to characterize how coregulation is concurrently and prospectively associated with symptoms of psychosis.

Social Disconnection in Young People at CHR for Psychosis

Adolescence and young adulthood have long been known as periods of upheaval in social relationships, with young people needing to navigate a changing social landscape (Blakemore, 2008; Blakemore & Mills, 2014). Effective interpersonal functioning in these novel situations depends on higher order social skills that develop during adolescence and into young adulthood (Blakemore, 2008). Yet, at the very time that typically developing young people show graded improvements in social cognition (Blakemore, 2008), those at risk for psychosis begin to exhibit prodromal symptoms that interfere with their interpersonal functioning (Williams et al., 2022).

Social impairments are a hallmark of psychosis spectrum disorders and involve altered social cognition, behavior, and network formation (Green et al., 2018). Crucially, these impairments emerge prior to disorder onset among those at CHR for psychosis (Cornblatt et al., 2012; Kimhy et al., 2016; Lincoln et al., 2017). At heightened risk for imminently developing a psychotic disorder, CHR individuals display subtle attenuated positive symptoms (e.g., brief, poorly formed hallucinations). The social difficulties evident in this population are frequently cited as a key concern (Lencz et al., 2004), are strongly associated with overall quality of life (Addington & Addington, 2008), and enhance risk for conversion to psychosis (Cannon et al., 2008). Underpinning these social difficulties, CHR individuals have social cognitive deficits that range from lower level processes, such as impaired emotion recognition (Allott et al., 2014; Kohler et al., 2010; Pelletier et al., 2013), to higher level complex social cognition (e.g., paradoxical intentions; Lincoln et al., 2017). Interpersonal behavior is also affected: CHR individuals show a tendency toward social withdrawal, blunted emotional expressivity, coldness, unusual behaviors, and hostility (Fattal et al., 2024; Glenthøj et al., 2020; Gupta, Osborne, et al., 2023; Gupta et al., 2022). These impairments likely contribute to difficulties in forming healthy social networks, leading to lower quality of life and worse social functioning (Ryan et al., 2023).

It takes two to tango, and a growing body of work shows that CHR individuals not only experience impairments in social functioning and lower social support, but so to do their relationship partners (Schlosser et al., 2010). For example, mothers of CHR youth report lower levels of emotional bonding with their child (Yee et al., 2021). Further, during interactions, caregivers make more critical (Domínguez-Martínez et al., 2014) and fewer positive (Carol & Mittal, 2015) statements and express more negative emotions (McFarlane & Cook, 2007; O'Brien et al., 2006, 2009; Schlosser et al., 2010). Such disruptions in social functioning extend beyond the traditional child–parent dyadic contexts that have been of frequent focus: In dyads composed of CHR youth and caregivers more broadly conceived, caregivers show increased negative emotions when talking about the CHR youth (Gupta, Horton, et al., 2023). Taken together, these studies indicate that there are marked impairments in social functioning in CHR individuals and such difficulties affect both members of the dyad.

Coregulation Supports Mental Health

Coregulation describes bidirectional influences between two members of a dyad that support well-being and mental health (Butler & Randall, 2013). Implicated in healthy relationship functioning throughout the lifespan (in childhood; Davis et al., 2018; in adulthood; Helm et al., 2012; Nelson et al., 2017), coregulation—and related concepts like linkage (Chen et al., 2021) and synchrony (Davis et al., 2018)—span levels of analysis (Butler & Randall, 2013) that range from observable behavior (Feldman, 2006) to latent emotional states (Steele et al., 2014) to physiological processes (Ferrer & Helm, 2013). In the case of biological systems where each person has a homeostatic setpoint, cross-partner influences help both people return to their baseline (Ferrer & Helm, 2013; Steele et al., 2014).

Several theoretical frameworks consider the role of coregulation in well-being and health (such as Beckes & Coan, 2011; Butler, 2011; Chen et al., 2021; Haase, 2023; Levenson et al., 2013). These frameworks highlight the importance of shared emotions (e.g., Levenson & Gottman, 1983; Wells et al., 2022) and synchronization of behavior—such as shifts in posture, gesture, gaze, and attention (Wells et al., 2022)—in physiological coregulation (Chen et al., 2021). In turn, physiological coregulation is thought to play a critical role in empathy, interpersonal functioning, and emotion regulation (Beckes & Coan, 2011; Butler & Randall, 2013; Levenson & Ruef, 1992; Preston & de Waal, 2002).

Whereas coregulation is frequently linked to adaptive outcomes, disruptions to coregulation are often implicated in relationship difficulties and mental health symptoms.[1] During childhood, failures in coregulation are associated with increasing externalizing and internalizing psychiatric symptoms (Hollenstein et al., 2004; E. Lunkenheimer et al., 2016). These effects extend into adulthood (e.g., Levenson & Gottman, 1983). For example, in romantic couples oversampled for interpersonal difficulties, disruptions to coregulation heighten discord and predict decreases in relationship satisfaction 1 year later (Schreiber et al., 2020). Interestingly, the foundational research on physiological coregulation was conducted

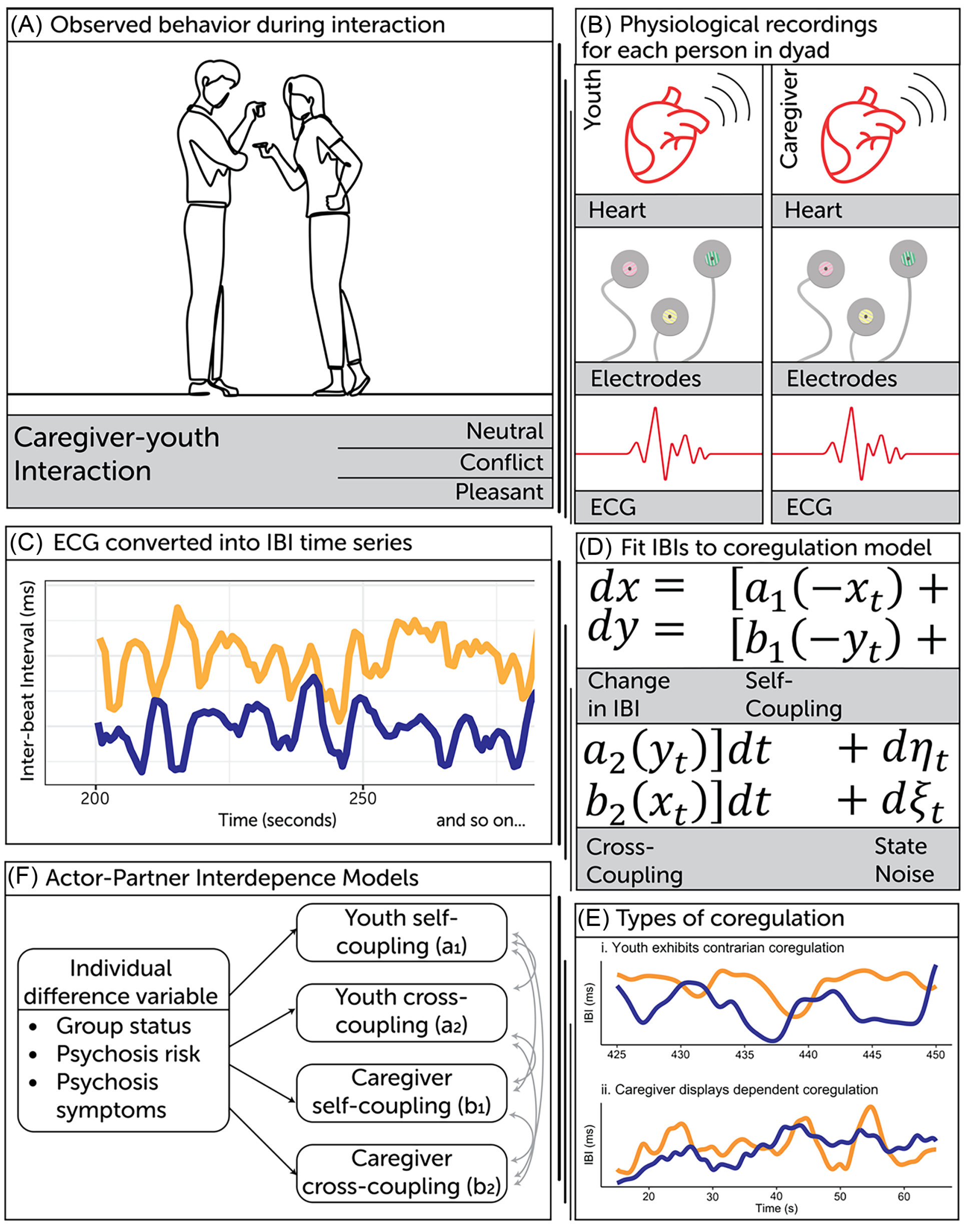
[1] Whether coregulation is linked to adaptive or maladaptive outcomes depends on context (Helm et al., 2012), the physiological system (Blasberg et al., 2023; Timmons et al., 2015), and analytic techniques (Butler, 2011; Chen et al., 2021).

in psychotherapy–client dyads (e.g., Dimascio et al., 1957), underscoring the clinical relevance of this construct.

The Present Study

We investigated the role of physiological coregulation in CHR individuals and their caregivers (study design summarized in Figure 1). Autonomic inflexibility is a transdiagnostic risk factor for psychopathology (Beauchaine, 2001; Beauchaine et al., 2007), is indexed by tonic high frequency heart rate variability (HF-HRV) or HF-HRV reactivity to stress (Beauchaine, 2015), and may undermine coregulation (Diamond & Hicks, 2005). Reduced HF-HRV is implicated in psychosis (Cacciotti-Saija et al., 2018; Clamor et al., 2016) and autonomic inflexibility has even been

Figure 1
Study Design



Note. (A) Overview of study session. (B) Description of data collection procedures. (C) Preprocessed ECG data converted to IBI time series. (D) Stochastic differential equation model of changes in heart rate to capture within-dyad process. (E) Representative IBI time courses of dyads exhibiting (i) contrarian and (ii) dependent coregulation. Purple = youth. Yellow = caregiver. (F) APIM analyses to characterize between-dyad level correlates of coregulation. ECG = electrocardiography; APIM = actor–partner interdependence model; IBI = interbeat interval. See the online article for the color version of this figure.

observed in CHR individuals (Kocsis et al., 2020; Latvala et al., 2016). Coherence between heart rate and facial expressions of emotion have also been shown to be disrupted in CHR individuals (Fattal et al., 2024). In light of this work, we focused our analyses on coregulation of *heart rate*, a physiological index that is sensitive to autonomic functioning (Butler & Randall, 2013) and is well-suited for capturing physiological coregulation (Chen et al., 2021). Coregulation of heart rate thus describes the tendency for two people who are *in sync* to show similarly timed decelerations and accelerations in heart rate. Further, when emotions are heightened, this synchronization can actually support both people's heart rate returning to baseline (Butler, 2011).

For each member of the dyad, we obtained *self-coupling* and *cross-coupling* parameter estimates, which index within-person and cross-partner influences in heart rate, respectively. In quantifying cross-partner influences, cross-coupling parameter estimates index coregulation. We term negative cross-coupling *contrarian* coregulation, consistent with prior work (Steele et al., 2014). Because contrarian coregulation has been implicated in many psychiatric conditions and interpersonal difficulties (Fischer et al., 2017; E. S. Lunkenheimer et al., 2011; Schreiber et al., 2020), we anticipated that psychosis risk would be related to contrarian coregulation—particularly during conflict interactions (Sbarra & Hazan, 2008; Timmons et al., 2015). In exploratory analyses, we examined whether contrarian coregulation predicted a worsening clinical course over the ensuing 12 months.

Method

Participants

Participants were recruited via email, social media advertising, newspaper advertising, flyers, and community health referrals as part of the Northwestern University Dyadic Interaction study from 2018 to 2022 (see Supplemental Material, for additional details). Participants were excluded if they were not between 12 and 30 years of age, had a history of head injury with loss of consciousness longer than 2 min, met criteria for a psychotic disorder, or if they had a neurological condition. In their initial visit, participants underwent structured clinical interviews (see the Clinical Assessments section) to identify whether a CHR syndrome was met. If no CHR syndrome was met, healthy control (HC) participants were excluded if they met criteria for any *Diagnostic and Statistical Manual of Mental Disorders*, fifth edition diagnosis other than depression and anxiety or if they had any family history of a psychotic disorder. We decided to enroll youth with (current or lifetime) depressive and anxiety disorders into the HC group so that the control group was more representative of the general population. These exclusion criteria only applied to the youth participants; caregiver participants were not excluded based on psychopathology.

The present study focused on dyads who jointly participated in interactions and for whom physiology data were available for both partners.[2] We refer to the target participant (CHR or HC) as "youth" since these participants were adolescents and young adults. We refer to the partner of the target as "caregiver." Our use of the term caregiver encompasses more traditionally defined caregivers, such as parents, and close others who often provide emotional support (e.g., romantic partners). The final sample consisted of 137 participants (32 CHR youth, 38 HC youth, 31 CHR caregivers, and 36 HC caregivers). For one CHR and two HC caregivers, the same caregiver interacted with two youth. This situation typically occurred when a parent had two children participate in the study. Two CHR youth were taking antipsychotic medications at the time of the visit. Analyses were rerun excluding those taking antipsychotics (detailed in Supplemental Material). Sociodemographic characteristics for youth and caregivers are presented in Table 1. All participants gave informed consent (or assent, if under age 18) prior to participating in the study.

Procedure

Youth and caregivers participated in two visits. In the first visit, participants completed clinical interviews and questionnaires. In the second, participants completed a laboratory-based dyadic interaction visit. Following established procedures in biobehavioral research with dyads (e.g., Levenson, 2024), youth and caregivers engaged in three 10-min videotaped conversations on the following: (a) events of the day (neutral conversation), (b) an issue of ongoing disagreement in their relationship with the instruction to come to a resolution (conflict conversation), and (c) a topic of mutual enjoyment (pleasant conversation). Electrocardiogram recordings were continuously collected during the three conversations. The Northwestern University Institutional Review Board approved all study protocol and procedures. Additional information about study procedures, including how procedures were adapted for the COVID-19 pandemic, can be found in the Supplemental Material.

Clinical Assessments

Interviews and assessments were conducted by trained doctoral students and postdoctoral professionals. To assess for psychosis-risk symptoms and to determine CHR status, the Structured Interview for Psychosis-Risk Syndromes (SIPS) Version 5.6.1 was administered. Raters were certified using official SIPS training developed by the creators of the scale. Total scores from the SIPS negative, positive, and disorganized symptom domains were used as indicators of their respective symptom dimensions. Affective symptoms were measured using the Beck Depression Inventory (Beck et al., 1961) and the Beck Anxiety Inventory (Beck et al., 1988). The Structured Interview for the *Diagnostic and Statistical Manual of Mental Disorders*, fifth edition, Research Version (SCID-5-RV) was used to determine the presence of mental health disorders (First et al., 2015). See Supplemental Tables S1–S3 for a diagnostic characterization of the sample, as well as information on symptom data and medication exposure. Although clinical information was collected on caregivers (e.g., Beck Depression Inventory, Beck Anxiety Inventory, SCID-5-RV), for the purpose of the present study, we only focus on the youth's symptomatology.

Psychosis risk was indexed using the North American Prodrome Longitudinal Study risk calculator (NAPLS-RC), which estimates an individual's probabilistic risk of developing a psychosis spectrum disorder over the course of 2 years (Cannon et al., 2016). The NAPLS-RC uses SIPS ratings and participant age, as well as other measures that were administered: the Brief Assessment of Cognition in Schizophrenia (Keefe et al., 2004), the Hopkins Verbal Learning

[2] In contrast to other work published using data from the Northwestern University Dyadic Interaction study (e.g., Fattal et al., 2024), which did not require complete physiology data for both members of the dyad.

Table 1
Demographic Characteristics of Sample

Demographic characteristic	Youth CHR (N = 32)	Youth HC (N = 38)	Caregiver CHR (N = 31)	Caregiver HC (N = 36)
(a) Person-level characteristic				
Average age	21.06 (3.14)	16.82 (3.62)	42.97 (16.58)	50.42 (7.99)
Race/ethnicity				
African American/Black	10	4	9	6
Asian	3	3	1	1
Caucasian/White	12	20	15	25
Latino/Hispanic	6	3	4	2
Multiracial	1	8	2	2
Gender				
Female	11	20	19	30
Male	19	18	11	6
Other	2	0	1	0

Demographic characteristic	CHR caregiver (N = 31)	HC caregiver (N = 36)
(b) Dyad-level characteristic		
Income		
<$20,000	1	3
<$35,000	8	1
<$50,000	4	3
<$75,000	4	5
<$100,000	2	4
<$150,000	3	8
>$150,000	9	12
Education		
<High school diploma/GED	2	0
High school diploma/GED	6	3
Associate's degree	4	2
Bachelor's degree	10	10
Graduate degree	9	21
Type of caregiver		
Parent	16	34
Relative	7	0
Romantic partner	8	1
Other	0	1

Note. Because dyad level characteristics reflect facets of the caregiver (e.g., income), the N in (b) is reflective of the total number of caregivers in the final sample. This N is slightly lower than that of youth, since there were a few instances in which the same caregiver was part of multiple dyads (e.g., a parent has two kids and both children participate in the study). For the one HC dyad with an "other" caregiver, the caregiver chose not to disclose additional information about the type of relationship. CHR = clinical high risk; HC = healthy control; GED = general educational development.

Test–Revised (Brandt & Benedict, 2001), negative life events scores from the Research Interview Life Events Scales (Dohrenwend et al., 1978), and social functioning decline as measured by the Global Functioning Scale (Carrión et al., 2016).

Electrocardiography

Electrocardiography (ECG) was measured using Mindware's BioNex 8-slot chassis and recorded using Biolab Acquisition Software (Mindware Technologies LTD, Gahanna, OH). Placement of electrodes followed standard recommendations from MindWare Technologies LTD. Pregelled medical ECG electrodes were placed in a lead II, configuration, with one electrode placed on the bottom-left side of the ribcage and a second on the right collar bone. The ground for the lead was placed on the bottom-right rib. Raw data were sampled at 1,000 Hz and were analyzed for artifacts by trained research assistants using MindWare HRV Analysis 3.1.5 and corrected if necessary, then averaged to produce a single value per second.

Analyses

Estimating Coregulation

Using stochastic differential equations, we jointly modeled each dyad member's heart rate as a function of their own heart rate and their partner's heart rate (Figure 1D). Interpersonal influences were indexed by a *cross-coupling* parameter and intrapersonal influences were captured with a *self-coupling* parameter. Models were implemented in a nonlinear stochastic state-space framework using the variational Bayesian analysis toolbox in Matlab (Daunizeau et al., 2014; Mathworks, 2020). We modeled coregulation for each dyad and for each dyadic context. After ensuring adequate fit (determined using model R^2), we extracted the posterior mean for each parameter.

Building on simulation studies of this model (Steele et al., 2014) and reminiscent of early work characterizing these dynamics in psychotherapist–client dyads (Dimascio et al., 1957), we term negative (<0) cross-coupling parameter estimates contrarian coregulation (conceptually related to antiphase linkage or negative cross-correlations in physiological systems; Chen et al., 2021). Simulation studies show (Steele et al., 2014) and empirical data confirm (Schreiber et al., 2020) that negative cross-coupling results in misalignment of heart rate, destabilizes the dynamical system, and undermines coregulation. Conversely, we term positive cross-coupling dependent coregulation (Steele et al., 2014), which is conceptually similar to synchrony (Butler, 2011; Davis et al., 2018) or in-phase linkage (Chen et al., 2021). Dependent coregulation "opens" an individual up to the influence of their partner and could allow them to receive emotional support. Importantly, despite the connotation of these terms, whether contrarian coregulation is maladaptive depends on the configuration and strength of the dyad's cross-coupling parameters (Steele et al., 2014), on the physiological system examined (Saxbe & Repetti, 2010), and on contextual factors (Helm et al., 2012).

Supplemental Table S4 provides an overview of key terms, particularly the mapping between psychological constructs (coregulation) and the operationalization of the construct (cross-coupling). When reporting results, we typically use terms that reflect the model's operationalization of the corresponding psychological constructs (i.e., cross-coupling). When interpreting results, we generally refer to the psychological construct (i.e., coregulation). To clarify whether contrarian coregulation was maladaptive in our sample, we conducted sensitivity analyses linking coregulation parameter estimates with social functioning.

Linking Coregulation With Psychosis Risk

Using actor–partner interdependence models (APIMs), we tested whether relevant diagnostic characteristics—for example, psychosis risk—were associated with coregulation parameter estimates. We fit APIMs using Bayesian structural equation models that were implemented in *Mplus* (B. Muthén & Asparouhov, 2012; L. K. Muthén & Muthén, 2017). Prior to interpreting model results, we ensured good global model fit (posterior predictive $p > .05$). We report standardized effects and their corresponding Bayesian two-tailed p values (Makowski et al., 2019). We focus interpretation on effects that are markedly different from zero ($p < .05$). All results, inclusive of those we did not interpret, are reported in Supplemental Tables S6–S15. For primary results, we report standardized effects (β) and the 95% credible intervals for these effects (denoted using square brackets).

Our analyses proceeded as follows. First, across all three conversations, we examined whether clinical group status predicted coregulation. Second, we tested whether psychosis-risk scores, as assed via the NAPLS-RC, were related to coregulation. Third, we examined whether specific types of psychosis symptoms predicted coregulation. Fourth, we assessed whether the effects of psychosis symptoms on coregulation were only found in the high-risk group. Specifically, we tested whether group status moderated the effect of psychosis symptoms on coregulation. When we found evidence of moderation, we plotted the effects of psychosis symptoms on coregulation in the CHR group. When we did not find evidence of moderation, we examined effects of symptoms for each group separately.

Exploratory Analyses

To understand whether disruptions in coregulation contribute to the developmental course of psychosis, we obtained 12-month follow-up data on a subset of individuals in the high-risk group ($N = 28$). We tested whether coregulation moderated the effect of baseline symptoms on psychosis symptoms 12 months later. Because of the small sample size, these analyses could not be conducted within an APIM framework and we interpret these results cautiously.

Assessing Sensitivity and Specificity

We conducted secondary analyses to determine whether effects were specific to psychosis risk or reflective of other commonly co-occurring psychiatric symptoms. To contextualize our results within the broader literature on coregulation and relationship health, we assessed whether coregulation was related to alterations in social functioning. We also conducted sensitivity analyses to ensure our results were robust to sociodemographic factors, including age, sex, race, income, and to model specification. We conducted sensitivity analyses to ensure our results were robust to relationship type and to medication exposure. Finally, we assessed whether results held when accounting for whether the session visit occurred pre- or post-COVID-19 onset. See Supplemental Material for details. Briefly, because CHR youth and HC youth differed in age, we only interpret effects that held when accounting for age.

Transparency and Openness

We follow American Psychological Association Journal Article Reporting Standards (Appelbaum et al., 2018). We report on all data exclusions and manipulations and on all study materials. The present research draws from a laboratory-based dyadic interaction study. Some results of this study have been reported previously (Fattal et al., 2024), but no prior study has examined physiological coregulation. As this study was part of a broader research project, we did not conduct a priori power analyses that were specific to this study. Nonetheless, in sensitivity analyses, we calculated the minimum effect size we were well-powered (80%) to detect (moderate effect sizes; see Supplemental Material, for further details). We used all available data from the broader research project for the purpose of this study. Data and materials for this study are not openly available, but interested parties may request accessing by contacting either corresponding author. Data analysis code is publicly available and can be found at https://github.com/alisonmarie526/chrp-coreg-analyses. We report on all data processing and analysis software in the corresponding Analyses section. This study's design and analysis were not preregistered.

Results

CHR Youth Exhibited Contrarian Coregulation

We first examined whether CHR youth showed greater contrarian coregulation (reflected in negative cross-coupling parameter estimates). During the conflict conversation, youth in the clinical high-risk group indeed evidenced negative cross-coupling (β = −0.35, [−0.55, −0.12], p = .004) and their caregivers exhibited elevated cross-coupling (β = 0.26, [0.02, 0.48], p = .032), relative to HC dyads.[3] While we did not find evidence that CHR group status was associated with cross-coupling during the pleasant conversation, we found robust associations for the neutral conversation: Youth in the high-risk group exhibited negative cross-coupling during the neutral conversation (β = −0.32, [−0.51, −0.10], p = .006; see also Supplemental Table S5 and Supplemental Figure S5).

Psychosis Risk Was Associated With Contrarian Coregulation

Psychosis risk, as measured via the NAPLS-RC, predicted lower cross-coupling in youth during the conflict conversation (β = −0.31, [−0.52, −0.06], p = .016). Though self-regulation was of less focal interest in the present study, psychosis risk was also associated with elevated self-coupling in the caregiver during the conflict (β = 0.33, [0.08, 0.53], p = .008) and pleasant (β = −0.34, [0.10, 0.54], p = .006) conversations. Altogether, these results indicate that psychosis risk is associated with contrarian coregulation, particularly during the conflict conversation.

Positive Symptoms of Psychosis Were Consistently Linked to Contrarian Coregulation

Positive symptoms predicted lower cross-coupling in the youth during the conflict (β = −0.29, [−0.48, −0.09], p = .006), pleasant (β = −0.27, [−0.47, −0.05], p = .016), and neutral (β = −0.24, [−0.44, −0.04], p = .020) conversations. We also found evidence that negative and disorganized symptoms were associated with lower cross-coupling, though these associations were less consistent: Whereas negative symptoms predicted lower cross-coupling in the pleasant (β = −0.38, [−0.57, −0.15], p = .002) and neutral (β = −0.21, [−0.42, −0.01], p = .040) conversations, disorganized symptoms were associated with lower cross-coupling in the pleasant conversation (β = −0.30, [−0.53, −0.03], p = .030).

Were Effects of Psychosis Symptoms on Coregulation Only Found in the High-Risk Group?

We next assessed whether the observed effects of psychosis symptoms on cross-coupling were moderated by group status. Among the observed associations that we report on above, only two were moderated by group status: As shown in Figure 2, the tendency for positive (β = −0.77, [−1.26, −0.23], p = .006) and negative (β = −0.89, [−1.42, −0.30], p = .004) symptoms to predict lower cross-coupling—and indeed negative cross-coupling—during the neutral conversation was driven by those in the clinical high-risk group. Given that we may have been underpowered to detect moderation, we also examined the remaining effects in the CHR and HC groups separately.

Across the moderation and group-specific analyses, a couple of patterns emerged: Our observation of negative cross-coupling during the conflict conversation in youth with more positive symptoms is driven by overall group differences. That is, effects of positive symptoms on coregulation during the conflict conversation were not significant when looking at effects within each group (ps > .68). Similarly, the negative cross-coupling observed in the pleasant conversation among youth with elevated disorganized symptoms is also likely driven by overall group differences (ps > .13). Conversely, among those in the high-risk group, greater levels of positive (β = −0.35, [−0.62, −0.02], p = .038) and negative (β = −0.40, [−0.67, −0.03], p = .034) symptoms predicted lower cross-coupling during the pleasant conversation. Taken together, these results show that the conflict conversation consistently elicited stable group differences in coregulation. In contrast, pleasant and neutral conversations provided insight into the subtle variations in patterns of physiological attunement between youth and their caregivers.

A Dynamic of Contrarian-Dependent Coregulation Exacerbated Positive Symptoms of Psychosis Over 1-Year Follow-Up

Youth (β_{pos} = −1.26, p = .030; β_{neu} = −2.42, p = .028) and caregiver (β_{pos} = 2.22, p = .011; β_{neu} = 3.85, p = .008) cross-coupling during the pleasant and neutral conversations moderated changes in positive symptoms over a 12-month follow-up. The dyads that were pulled into a contrarian-dependent dynamic (i.e., negative youth cross-coupling and positive caregiver cross-coupling; Supplemental Table S4) during these interactions were the same dyads in which the youth experienced worsening positive symptoms of psychosis over the course of 1 year. We found a similar effect for the conflict conversation: caregivers who exhibited elevated cross-coupling tended to be in dyads where the youth exhibited worsening positive symptoms at the 12-month follow-up (β = 2.67, p = .035).

Discussion

Physiological coregulation plays an important role in socioemotional development (Davis et al., 2018; Feldman, 2006; Harrist & Waugh, 2002). Failures in coregulation in turn have been implicated in the development of psychopathology (E. S. Lunkenheimer et al., 2011). The present dyadic interaction study sought to probe—for the first time to our knowledge—alterations in physiological coregulation among dyads composed of CHR individuals and their caregivers.

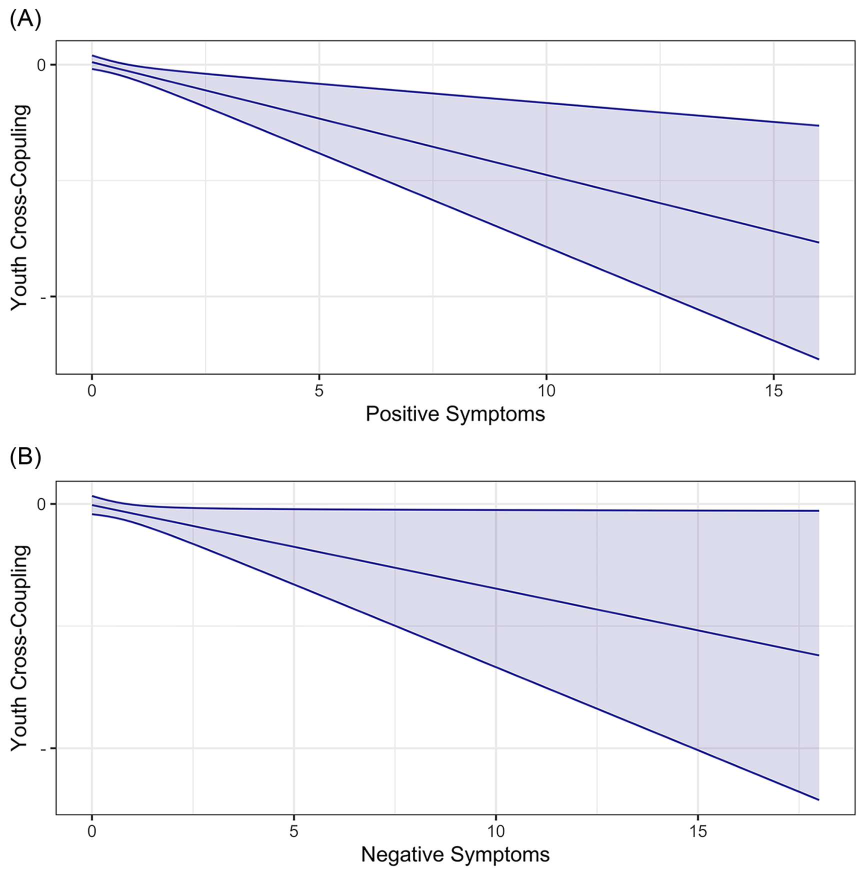
Contrarian Coregulation in Young People at Risk for Developing Psychosis

Consistent with our hypotheses, those at CHR for psychosis exhibited contrarian coregulation with their caregiver during a conflict conversation. That is, as they talked about an area of disagreement in their relationship, when their caregiver's heart rate slowed, the CHR individual's heart rate hastened, and vice versa. This finding builds on prior work showing that this contrarian dynamic escalates negative emotions (Steele et al., 2014), is associated with tension in psychotherapy sessions (Marci & Orr, 2006; Marci et al., 2007), and is implicated in deteriorating relationship satisfaction (Hilpert et al., 2023; Schreiber et al., 2020).

[3] Group-level means of coregulation parameter estimates support this interpretation (see Supplemental Table S5 and Supplemental Figures S3–S4).

Figure 2

Positive and Negative Psychosis Symptoms Predict Contrarian Coregulation During Neutral Interaction Among High-Risk Youth



Note. (A) Effect of positive symptoms on coregulation. (B) Effect of negative symptoms on coregulation. Because the upper bound of model-predicted cross-coupling values is zero, lower cross-coupling is indicative of contrarian coregulation. See the online article for the color version of this figure.

To contextualize our results (Helm et al., 2012), we examined whether coregulation was *also* disrupted during interactions in neutral and pleasant conversations. Even in these contexts that may not be perceived as interpersonally challenging, we found evidence of disrupted coregulation: (a) CHR individuals exhibited contrarian coregulation during a neutral conversation and (b) positive and negative symptoms of psychosis were associated with contrarian coregulation during the neutral and pleasant conversations. In sensitivity analyses, we found that contrarian coregulation during these interactions was associated with worse social functioning. Interestingly, neutral and pleasant conversations appeared to be especially sensitive to symptom severity, rather than overall group differences. These contexts may function as a "weak" rather than "strong" situation (Hedge et al., 2018) and be useful for understanding gradations in social functioning and psychosis symptoms.

Importantly, we further found that CHR individuals who exhibited contrarian coregulation experienced worsening positive symptoms of psychosis over the course of 1 year. This was particularly true when the caregiver also showed dependent coregulation. Our results suggest that disruptions to coregulation are not simply the *consequence* of a disease process. Instead, contrarian coregulation is a physiological index of social dysfunction and may even contribute to the course of the disorder.

Physiological Coregulation in Psychopathology

The psychosis prodrome—including associated social impairments—emerges during adolescence and young adulthood (Cornblatt et al., 2012), times of major transformation in social, emotional, and biological functioning (Andrews et al., 2021; Blakemore, 2008; Blakemore & Mills, 2014). Our finding that physiological coregulation is linked to poorer social functioning and is disrupted in those at risk for developing psychosis aligns with a broader literature on social difficulties in this population: CHR individuals show declines in social functioning (Cannon et al., 2008) that are accompanied by deficits in lower level (Allott et al., 2014; Kohler et al., 2010; Pelletier et al., 2013) and complex social cognitive processes (Lincoln et al., 2017). Blunted expressions of emotions, particularly positive emotions, and alterations in interpersonal behaviors likely also contribute to interpersonal difficulties (Fattal et al., 2024; Glenthøj et al., 2020; Gupta, Osborne, et al., 2023; Gupta et al., 2022; Williams et al., 2022). These altered social behaviors may help to partly explain why CHR youth exhibited contrarian coregulation with their caregiver: reduced social cognition and blunted emotional expressions could make it challenging for the CHR individual to get *in sync* with their caregiver.

Yet, social impairments are not specific to psychosis. Social difficulties can be part of a psychiatric disorder (e.g., autism spectrum disorder) and can be a consequence of psychopathology (e.g., social isolation in depression; Davila et al., 1997). Consistent with our primary findings, we have previously shown that physiological coregulation is disrupted in psychiatric disorders typified by interpersonal difficulties (e.g., personality disorders; Schreiber et al., 2020). Other studies have further implicated failures in coregulation in the development of internalizing and externalizing psychopathology (Hollenstein et al., 2004; E. Lunkenheimer et al., 2016). Indeed, even in our present study where we oversampled for psychosis, internalizing symptoms were associated with contrarian coregulation during the pleasant conversation (see Supplemental Material).

Thus, disruptions to physiological coregulation may be implicated in a wide variety of psychiatric disorders. Coregulation plays an important role in emotional development (Davis et al., 2018; Harrist & Waugh, 2002) and emotion dysregulation is not only a core feature of many disorders (e.g., borderline personality disorder; American Psychiatric Association, 2013) but is broadly implicated in the development of psychopathology (Beauchaine et al., 2007; Cicchetti et al., 1995). Our findings and the broader literature underscore the role of healthy relationship functioning in emotional well-being and mental health. Whereas healthy relationships support well-being (Holt-Lunstad et al., 2010) and protect against psychopathology (Whisman et al., 2000), the converse also holds true. Disruptions to social processes—like coregulation (Butler & Randall, 2013)—can lead to myriad functional outcomes, including relational difficulties (Frith & Frith, 2007), heightened risk for psychopathology (Decety & Moriguchi, 2007; Whisman, 2007), exacerbation of existing social difficulties, and a worsening psychiatric course (Whisman & Uebelacker, 2009).

Strengths and Limitations

Our novel study design allowed us to characterize coregulation across *multiple* dyadic contexts, including those that are theoretically central—such as a conflict discussion (Sbarra & Hazan, 2008; Timmons et al., 2015)—and less central to the construct (Ferrer & Helm, 2013). However, the order of dyadic contexts was not randomized, consistent with established procedures (e.g., Levenson, 2024), and we cannot rule out the possibility of carryover effects. While this limitation could be overcome with counterbalancing, we note that ethical considerations—like minimizing end-of-session participant distress—may preclude randomization.

Our sampling frame meant we were well-poised to understand the role of physiological coregulation in the *development* of psychosis. Despite this strength, we note three limitations. First, the sample size was relatively small for APIMs. This concern is especially true for our exploratory analyses investigating changes in psychosis symptoms over a 12-month follow-up, which could only be conducted on a subsample. To mitigate this concern, we used a Bayesian estimator that is more appropriate for small samples (B. Muthén & Asparouhov, 2012). Further, sensitivity analyses verified all primary results held in simpler regression models, which led us to conclude our results were robust to model specification. Future research using a larger sample is nonetheless warranted. Second, we employed a case–control design. Although there are advantages to this approach (e.g., power), there are several disadvantages (Preacher et al., 2005). Our use of extreme groups may have reduced variability on continuous variables (e.g., within group variability in psychosis-risk symptoms) and could have hindered those analyses. Third, although cases and controls were matched on key demographic variables (e.g., sex and race; $ps > .14$), there were group differences in other demographic factors, including income and age. Further, parent–child dyads were less common in the CHR group. In sensitivity analyses, we confirmed that group differences in coregulation parameter estimates were not better explained by these demographic characteristics.

We also considered the possibility that medication exposure could have impacted our results. Due to data availability, we could not directly assess the role of medications that directly alter heart rate variability. Nonetheless, sensitivity analyses confirmed our results were robust to overall medication exposure, psychotropic medication exposure, and psychotropic medication class. Many factors that influence heart rate variability were not assessed in this study (e.g., phase of menstrual cycle, exercise). We were thus unable to assess whether our results held when controlling for all potentially relevant confounding variables. Finally, we limited the scope of the present study to focusing on how youth's psychosis symptoms affected coregulation with their caregiver. Future work examining how caregiver's symptoms impact coregulation is warranted.

Our use of state-of-the-art methods to capture physiological coregulation has several advantages. First, we quantified *directed* momentary interpersonal influences on heart rate (Ferrer & Helm, 2013). Second, relative to initial instantiations of the model (Steele et al., 2014), we modeled state noise attributable to random fluctuations in heart rate. Third, we fit the model to each dyad and for each conversation separately, ensuring that parameter estimates were not unduly regularized (Ferrer & Helm, 2013; Ferrer & Steele, 2012). However, our two-step approach precluded us from propagating uncertainty of the parameter estimates to between-person analyses.

Finally, our study focused on cross-partner influences in heart rate, yet there are many other forms of interpersonal influence that occur during an interaction (Butler & Randall, 2013). Indeed, coregulation of respiration (Helm et al., 2012), behavior (Feldman, 2006), and vocal frequency (Atkinson, 1978; Fischer et al., 2017) is commonly observed. Further complicating matters, coregulation in certain biological systems may actually lead to worse relationship functioning (e.g., cortisol; Saxbe & Repetti, 2010), though differences in how coregulation is computed may explain some discrepant findings (Chen et al., 2021). To aid interpretation of our findings, we assessed whether coregulation was related to broader social functioning in the sample. Similar to past research (Schreiber et al., 2020), contrarian coregulation was associated with worse social functioning. Nonetheless, future research examining concurrent, moment-to-moment behavioral indicators of coregulation (e.g., facial expressions) may provide crucial insights into *how* interpersonal processes get under the skin and the contextual factors that determine whether these interpersonal processes will support or hinder relationship health. Future work in CHR populations may also help clarify whether particular social deficits (e.g., blunted emotional expressions) drive contrarian coregulation and a worsening clinical course.

Constraints on Generality

Our sample was recruited from the general community and from mental health clinics in a midsized Midwestern city. Given sampling biases that can occur when using convenience samples and when recruiting within a single geographic region, our sample is not fully representative of the broader U.S. population, let alone the global population. Future research using more racially, socioeconomically, and geographically diverse samples is needed to ensure that our results generalize to these populations.

Conclusion

The present study showed that young people at risk for developing psychosis exhibited a contrarian-dependent coregulation dynamic with their caregiver, and contrarian coregulation was associated with worse social functioning. CHR individuals who displayed this dynamic with their caregiver during neutral and pleasant conversations evidenced worsening positive symptoms of psychosis over the course of the next year. These results align with a growing literature implicating physiological coregulation in social functioning and linking contrarian coregulation with psychopathology. Psychosocial interventions that target physiological coregulation may hold promise for improving social functioning in those at risk for psychosis and, in line with important theoretical and empirical frameworks (e.g., Beckes & Coan, 2011; Eisenberger, 2012; Feldman, 2006; Levenson, 2024), contribute to a biologically grounded science of social connection.

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